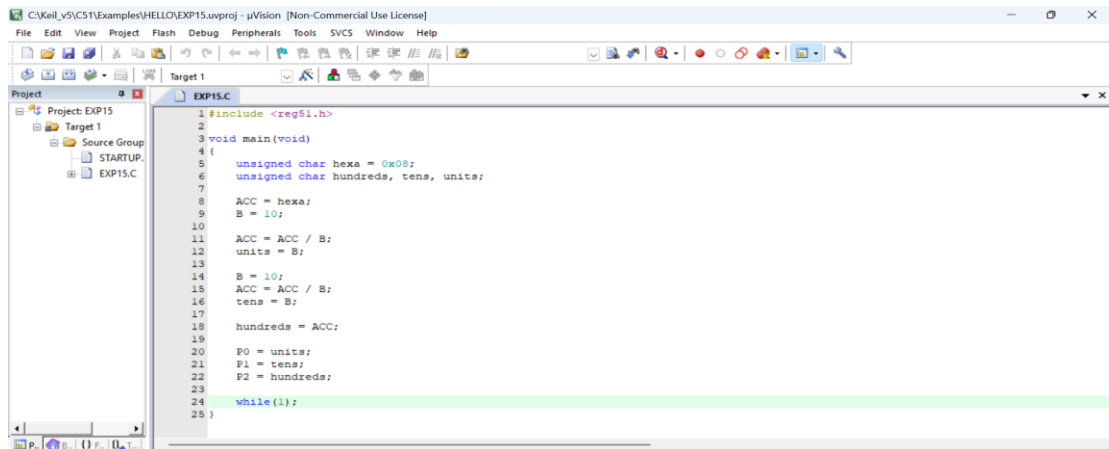
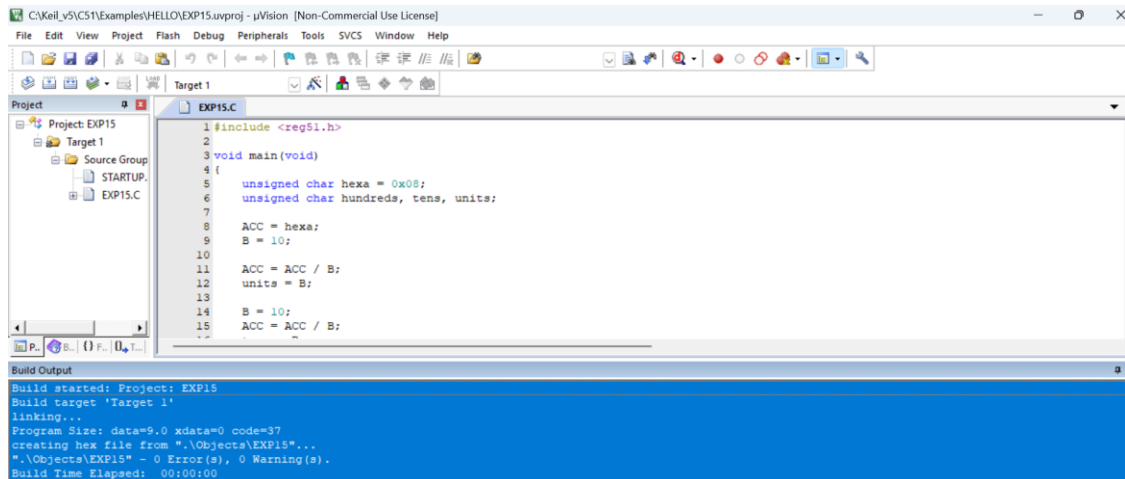


Exp No: 15

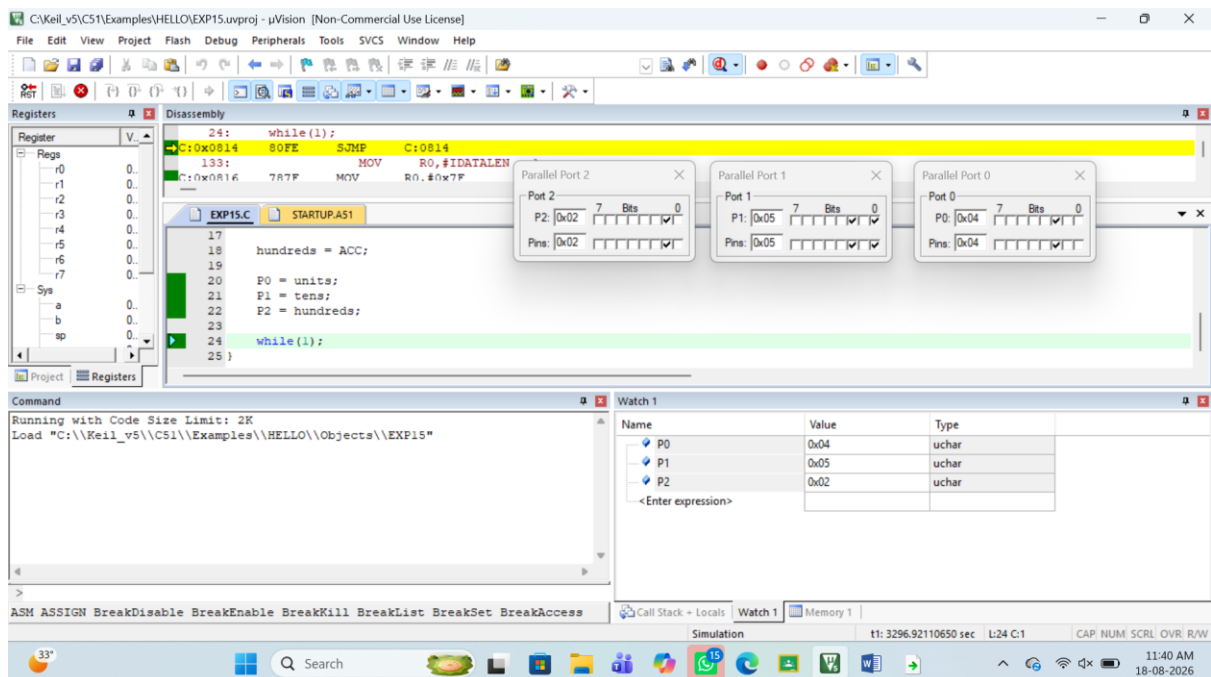
EMBEDDED C PROGRAM TO CONVERT THE HEXADECIMAL DATA 0XCFH TO DECIMAL AND DISPLAY THE DIGITS ON PORTS P0, P1 AND P2



```
1#include <reg51.h>
2
3void main(void)
4{
5    unsigned char hexa = 0x08;
6    unsigned char hundreds, tens, units;
7
8    ACC = hexa;
9    B = 10;
10
11    ACC = ACC / B;
12    units = B;
13
14    B = 10;
15    ACC = ACC / B;
16    tens = B;
17
18    hundreds = ACC;
19
20    P0 = units;
21    P1 = tens;
22    P2 = hundreds;
23
24    while(1);
25}
```



```
Build started: Project: EXP15
Build target 'Target 1'
linking...
Program Size: data=9.0 xdata=0 code=37
creating hex file from ".\\Objects\\EXP15"...
".\\Objects\\EXP15" - 0 Error(s), 0 Warning(s).
Build Time Elapsed: 00:00:00
```



Registers

Register	Value
r0	0x00
r1	0x00
r2	0x00
r3	0x00
r4	0x00
r5	0x00
r6	0x00
r7	0x00
SP	0x00

Disassembly

Address	Disassembly
0x0014	24: while(1): SJMP C:0814
0x0015	133: MOV R0, #IDATALEN
0x0016	787F: MOV R0, #0x7F

EXP15.C

```
17    hundreds = ACC;
18
19    P0 = units;
20    P1 = tens;
21    P2 = hundreds;
22
23    while(1);
24
25
```

Watch 1

Name	Value	Type
P0	0x04	uchar
P1	0x05	uchar
P2	0x02	uchar

Simulation

tt: 3296.92110650 sec L:24 C:1 CAP NUM SCRL OVR: R/W

11:40 AM 18-08-2026

